

Take-Away Games

Patterns that repeat, and why they must

Name: _____

How to play

- Two players share one pile of counters and take turns. On your turn you must take some counters; the **rule** says how many you may take. You can't take more than the pile has.
- Whoever takes the **last counter wins**.
- We write a rule in braces: **{1, 2}** means "take 1 or take 2."

Group of three? Two play and the third is the referee who checks every move. Rotate each game.

W piles and L piles

Always look at a pile from the point of view of the player **about to move**.

- **W pile**: the player to move can force a win, whatever the opponent does.
- **L pile** (a trap): the player to move will lose if the opponent plays well.

Working backward. 0 is L (your opponent just took the last counter). A pile is **W** if **at least one** move leads to an L pile. A pile is **L** if **every** move leads to a W pile.

Warm-up rule: {1, 2}

- 1 Play a few games starting with 10 or 11 counters. Who wins, the first or the second player?
- 2 Label each pile W or L for the rule **{1, 2}**.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
L															

- 3 Every multiple of 3 is an L pile. Explain why. Your explanation must work against **every** move your opponent could make. (What do you do after they take 1? After they take 2?)

- 4 You face a pile that is **not** a multiple of 3. What is your winning move?

A new rule: {1, 3, 4}

Take 1, 3, or 4 counters (not 2). With 2 counters left, the only move is to take 1.

5 Play a few games starting with 10 to 15 counters. Make a guess: which piles do you **not** want on your turn?

6 Label every pile from 0 to 29 as W or L. Work from small to large, using the rules for working backward.

0	1	2	3	4	5	6	7	8	9
L									
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29

7 List the L piles you found: _____

8 Describe the pattern. Can you say it using remainders?

A pile is L exactly when dividing it by leaves a remainder of or .

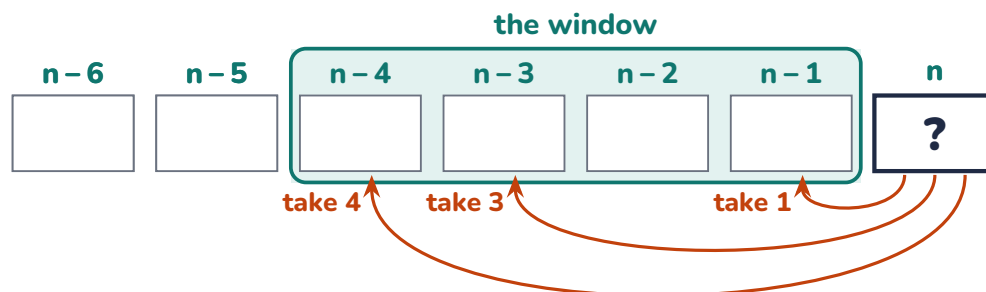
9 Predict W or L, and explain how you know.

50: 100: 1000: _____

10 The pile has 1000 counters and it is your turn. How many should you take? Why?

Why does the pattern repeat?

To label pile n under **{1, 3, 4}**, you only need the labels of piles $n-1$, $n-3$, and $n-4$. All three sit inside the **window** of 4 piles just before n .



- 11** Copy your labels from page 2 into these two windows.

0	1	2	3

piles 0 to 3

7	8	9	10

piles 7 to 10

What do you notice? _____

- 12** Suppose two windows have exactly the same labels. Explain why the label just after each window must be the same too, and then why the labels keep matching forever.

- 13** A window has 4 labels, each W or L. How many different windows are possible?

The pigeonhole principle

If more pigeons than holes fly into the holes, then some hole gets at least two pigeons.

- 14** Explain why **every** rule whose biggest move is 4 must have a win/lose pattern that eventually repeats. Hint: think about the windows that start at piles 0, 1, 2, ..., 16.

- 15** If the biggest move is 10, how many different windows are possible?

Change the rule

16 Label piles 0 to 19 as W or L for each rule. Then write how long the repeating block is.

{1, 2, 3}

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	L																			

repeating block length:

{1, 4}

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	L																			

repeating block length:

{1, 3, 5}

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	L																			

repeating block length:

{1, 2, 4}

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	L																			

repeating block length:

your rule:

{ }

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	L																			

repeating block length:

17 Under **{1, 3, 5}** the L piles are exactly the even numbers. Explain why. (Hint: every move takes an odd number.)

18 **{1, 2}** and **{1, 2, 4}** have the same L piles. Why doesn't the extra move "take 4" help anyone?

19 **Greedy Gus** always takes the biggest number the rule allows. Under **{1, 3, 4}**, find a pile where Gus's move loses, but a different move would have won.

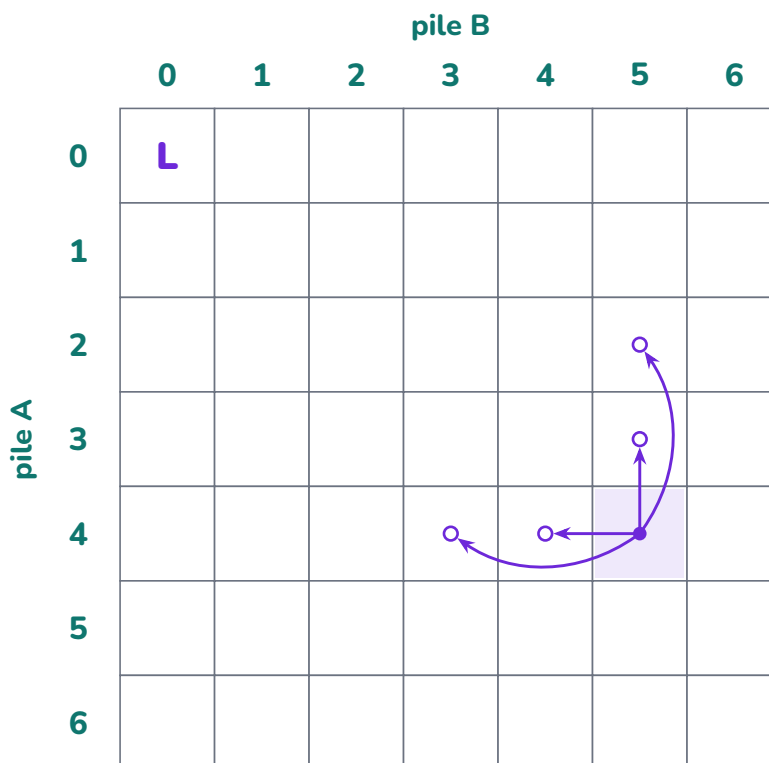
Pile: Gus takes A winning move is to take

Challenge: two piles

Two-pile rule

There are **two** piles. On your turn, choose **one** pile and take 1 or 2 counters from it. Whoever takes the very last counter wins.

20 Label each position W or L. The cell in row a and column b means “ a counters in one pile and b in the other.” A move goes 1 or 2 cells up, or 1 or 2 cells left (the arrows show the moves from the shaded cell, which means piles of 4 and 5).



21 Which positions are L? Describe the pattern. _____

22 Explain why two **equal** piles (like 5 and 5, or 100 and 100) are always L. (Hint: copycat.) _____

23 Without drawing a bigger grid: is the position with piles 14 and 20 W or L?

Challenge: beyond win and lose

Mathematicians give each pile a number called its **Grundy value**. It tells you more than just W or L.

Grundy values

The **mex** of a list of numbers is the **smallest** of 0, 1, 2, 3, ... that is **not** in the list. For example, mex of (0, 1, 3) is 2, and mex of (1, 2) is 0.

The **Grundy value** of a pile is the mex of the values of all the piles you can move to. A pile of 0 has value 0. A pile is L exactly when its value is 0.

- 24** Find the Grundy values for the rule **{1, 3, 4}**. Two are done for you. What pattern do you see?

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	1													

Example: from pile 1 you can only move to pile 0 (value 0). The mex of (0) is 1.

Pattern: _____

A fact about two piles

Play two piles that both use the same rule; on your turn, take from one pile only. The player about to move is in an **L** position exactly when the two piles have **equal** Grundy values.

- 25** Under **{1, 3, 4}**, piles of 4 and of 6 are both W piles on their own. What are their Grundy values? What happens if you play with **two** piles, 4 and 6? Check by playing!

- 26** Find a pile n that is not 5 such that two piles, 5 and n , make an L position.

Still unsolved

In some take-away games you may also split a pile into two piles. For a big family of games like these, called finite octal games, the mathematician Richard Guy guessed that the pattern of Grundy values always repeats in the end. Nobody has proved it yet, and nobody has found a game where it fails. It is a real open question!

To read more: Berlekamp, Conway, and Guy, *Winning Ways*; Siegel, *Combinatorial Game Theory*.